

Amadreza Farahani | ML/AI Software Engineer

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 [linkedin.com/in/amadrezafrh](https://www.linkedin.com/in/amadrezafrh) |  +39 351 641 1678 |  **+5 Years of Experience**

Skills

Agentic AI	LangGraph, LangChain, LangSmith, MCP, LlamaIndex, agentic RAG (hybrid retrieval, reranking), guardrails
LLM Systems	Transformers, HuggingFace, Ollama, vLLM, TorchServe, Milvus, ChromaDB, Pydantic, LoRa, QLoRa
Machine Learning	NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, MLflow, GANs, U-Net, YOLO, CNN, ASR
Cloud Technologies	AWS (Bedrock, ECS, EKS, EC2, S3, Lambda, SageMaker), Docker
Software Development	Python, C/C++, SQL, Bash, Linux, Git, CI/CD, FastAPI, Airflow, Design Patterns, DSA

Professional Experience

AI/ML Software Engineer

Zirak

Turin

11/2024 – Current

- Fine-tuned Llama 3.2 3B LLM with **LoRA/QLoRA** for an email-triage agent that classifies requests vs. non-requests and extracts structured fields via tool calling (LangGraph + Pydantic schemas), running fully on-premise along with ChromaDB for local vector storage.
- Created a production NL-to-SQL agent for **Poste Italiane's** logistics division using **LangGraph** with **MCP**-exposed tools, routing tool calls across PostgreSQL, and MySQL; FastAPI with Ollama for on-prem inference, cutting query authoring time by ~70% for non-technical analysts.
- Developed a cross-platform multi-agent meeting assistant (Electron/React + Python/FastAPI) combining real-time transcription, diarization, speaker verification, and agentic **RAG (LlamaIndex)** with hybrid retrieval and reranking (**LangChain/LangGraph**); LangSmith tracing and regression evals, reducing summarization hallucinations by ~25%.
- Deployed an LLM assistant for consultancy workflows, serving HuggingFace models via **vLLM/TorchServe** behind **FastAPI** with **Guardrails** and **LangSmith** for traceability; saved ~50 hours/month of manual document review, enabling the product for additional clients.
- Designed a custom U-Net for multispectral imagery with an **Airflow**-orchestrated (MWAA) ingestion, training and evaluation pipeline on AWS SageMaker, outperforming Random Forest, **XGBoost** and LightGBM baselines by ~10% F1 and reducing manual image review by ~80%.
- Built a rail-safety Computer Vision pipeline (YOLOv8/**PyTorch**) to detect sign defects and segment/track rails from frontal and nadir point of views, deployed to production on **AWS ECS**.

Visiting Researcher – Master Thesis

Technische Universität Darmstadt | CROSSING

Darmstadt

09/2023 – 11/2024

- Architected a Diffusion **Transformer** for voice conversion (Python, PyTorch) on HPC GPUs, improving speaker similarity by ~9% over open-source baselines with fewer sampling steps, enabling real-time conversion.
- Served low-latency inference on **AWS ECS** with **EC2** GPUs, using **MLflow** to track experiments and monitor performance.

Machine Learning Software Engineer

PartAI Research Center

Tehran

02/2020 – 03/2022

- Trained an on-device keyword-spotting Android app in **Java**, using hard negative samples to reduce false positives by ~30%.
- Optimized a speaker verification system on **TensorFlow**, fine-tuned on new domains that reduced the Equal Error Rate (EER) by ~2%.
- Established Test-Driven Development **TDD** for Persian **ASR** and **NLP** services; wrote unit/integration tests to reach 100% coverage and set up **Jenkins CI/CD** to automate builds, tests, and releases.
- Mentored new team members on clean, production-ready **Python** (testing, logging, design pattern), improving code quality and safety.
- Implemented state-of-the-art **Neural Networks** for speaker identification, optimizing for both cost-efficiency and operational safety.

Internship Experience

Associate Generative AI Engineer

AROL S.p.a

Canelli

09/2022 – 04/2023

- Shipped a synthetic machinery data generator (TensorFlow, trained on **AWS SageMaker** with **S3**) and a customer testing dashboard, packaged with **MongoDB** + Docker Compose (client/server/DB) and REST API docs so customers run it locally without exposing proprietary data.
- Defined KPIs and Python tests to validate synthetic vs. real machinery signals and API responses, integrating these checks into the team's **CI pipeline** and reducing manual QA effort for customer demos.
- Reviewed over 2,000 lines of production **Python** and ML pipeline code (training, inference, data preprocessing), adding tests and reducing compute cost while making models easier to debug and extend.

Computer Vision - Bachelor Thesis

PartAI Research Center | Tehran University of Science and Technology

Tehran

09/2019 – 02/2020

- Automated a traffic-flow monitoring system (**C++/Python**, **OpenCV**, YOLOv3, CNN) for vehicle detection, tracking, and flow estimation, reducing manual video review in a national-scale project.
- Engineered data analysis pipelines using **NumPy** and **Pandas** to structure annotation datasets, using **scikit-learn** to compute performance metrics and validate system reliability.

Education

M.Sc. Computer Engineering - Artificial Intelligence & Data Analytics

Politecnico di Torino

Turin

2021 – 2024

- Thesis:** RobVC: Robust Zero-Shot Self Supervised Voice Conversion (Funded by CROSSING Co. in Darmstadt) | [Blog](#)

B.Sc. Electrical and Computer Engineering - Telecommunications

Tehran University of Science and Technology

Tehran

2016 – 2021

- Thesis:** Deep Learning-Based Vehicle Detection and Traffic Flow Analysis (Funded by Megamouj Co.) | [Blog](#)